

On-the-fly Starting of a Squirrel-Cage Induction Motor in dq Synchronous Reference Frame

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Abstract—The report contains the mathematical modeling and simulation of a 30 kW squirrel-cage induction motor modeled in the synchronous (d - q) reference frame. The motor is initially running with its stator terminals open, and the rotor freewheeling in the positive direction. At the instant, when the rotor speed is at 120% of its rated value, the stator is connected to a balanced three-phase supply of rated magnitude and frequency. The resulting transient responses of stator currents, rotor speed and developed torque are recorded which show a negative torque acting on the machine reducing its speed till it settles to synchronous speed after connection.

Index Terms—Induction Motor, synchronous reference frame, dq model, rotor flux linkage, transient response

I. INTRODUCTION

Induction Machines are modeled in the synchronous reference frame to simplify transient analysis by converting all the time varying AC quantities into time invariant DC quantities. In this report, we have modeled such a case in MATLAB/Simulink for a 30 kW, 380 V, 1450 rpm, 50 Hz, squirrel-cage induction motor operating under super-synchronous conditions before its connection to the rated supply.

II. MATHEMATICAL ANALYSIS

We used a synchronous reference frame rotating at the electrical angular speed ω_s , where $\omega_s = 2\pi f$. The state variables considered are the stator currents i_{ds} , i_{qs} , i_{0s} and the rotor flux linkages ψ_{dr} , ψ_{qr} . The following constants are used in the subsequent sections $L_s = L_{lls} + L_m$, and $L_r = L_{llr'} + L_m$.

A. Conversion Matrices

We use the following matrix conversion to convert the given ABC -phase quantities into their corresponding $dq0$ quantities:

$$[X_{dq0}] = \frac{N_3}{N_2} \begin{bmatrix} \cos \delta & \sin \delta & 0 \\ -\sin \delta & \cos \delta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} [X_{abc}] \quad (1)$$

To re-obtain the original ABC quantities, the following matrix

conversion is used:

$$[X_{abc}] = \frac{2}{3} \frac{N_2}{N_3} \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \cos \delta & -\sin \delta & 0 \\ \sin \delta & \cos \delta & 0 \\ 0 & 0 & 1 \end{bmatrix} [X_{dq0}] \quad (2)$$

where, $\delta = \omega_s + \delta_0$ and X_{abc} , X_{dq0} refer to the voltage and current matrices.

B. Stator Current Equations

Consider the following constants to simplify our equations

$$\sigma = L_s - \frac{L_m^2}{L_r}, \text{ and } B = R_s + \frac{L_m^2}{L_r^2}$$

$$\frac{di_{ds}}{dt} = \frac{1}{\sigma} [V_{ds} - B \cdot i_{ds} + \sigma \cdot \omega_s i_{qs} + \frac{L_m}{L_r^2} R_r \psi_{dr} + \frac{L_m}{L_r} \omega_r \psi_{qr}] \quad (3)$$

$$\frac{di_{qs}}{dt} = \frac{1}{\sigma} [V_{qs} - \sigma \cdot \omega_s i_{ds} - B \cdot i_{qs} - \frac{L_m}{L_r} \omega_r \psi_{dr} + \frac{L_m}{L_r^2} R_r \psi_{qr}] \quad (4)$$

$$\frac{di_{0s}}{dt} = \frac{1}{L_{ls}} (V_{0s} - R_s i_{0s}) \quad (5)$$

C. Rotor Flux Linkage Equations

Consider the following constants to simplify our equations

$$\omega_{slip} = \omega_s - \omega_r$$

$$\frac{d\psi_{dr}}{dt} = L_m \frac{R_r}{L_r} i_{ds} - \frac{R_r}{L_r} \psi_{dr} + \omega_{slip} \psi_{qr} \quad (6)$$

$$\frac{d\psi_{qr}}{dt} = L_m \frac{R_r}{L_r} i_{qs} - \omega_{slip} \psi_{dr} - \frac{R_r}{L_r} \psi_{qr} \quad (7)$$

D. Electromagnetic Torque Equation

$$T_e = \left(\frac{P}{2}\right) \left(\frac{L_m}{L_r}\right) [\psi_{dr} i_{qs} - i_{ds} \psi_{qr}] \quad (8)$$

E. Fundamental Torque Equation

$$\frac{d\omega_r}{dt} = \left(\frac{1}{J}\right) [T_e - T_{load} - B\omega_r] \quad (9)$$

The ω_r obtained from this equation is expressed in mechanical radians per second, whereas the ω_r present in the equations discussed above is in electrical radians per second. To convert between the two, we use the following relation:

$$\omega_{relec} = \frac{P}{2} \omega_{rmech} \quad (10)$$

III. MODEL PARAMETERS

The following model parameters were considered:

- 1) Stator Configuration: *Star with Isolated Neutral*
- 2) Number of Poles: 4
- 3) Rated Power: 30 kW
- 4) Rated Voltage: 380 V
- 5) Frequency: 50 Hz
- 6) Rated Speed: 1450 rpm
- 7) Resistances: $R_s = R'_r = 0.1273\Omega$
- 8) Leakage Inductance: $L_{ls} = L'_{lr} = 1.341\text{ mH}$
- 9) Magnetizing Inductance: $L_m = 45.219\text{ mH}$
- 10) Rotor Moment of Inertia: $J = 1.6\text{ kg} \cdot \text{m}^2$
- 11) Friction and Windage Coefficient: $B = 0$
- 12) $\frac{N_3}{N_2} = \sqrt{\frac{2}{3}}$ (Power Invariance)

Note: The rotor parameters are referred to stator side.

IV. INITIAL CONDITIONS AND INPUTS

- 1) Initial Stator currents: $i_{ds}(0) = i_{qs}(0) = 0$
- 2) Initial Rotor fluxes: $\psi_{dr}(0) = \psi_{qr}(0) = 0$
- 3) Initial Rotor speed: $\omega_r(0) = 1.2 \times 1450 = 1740\text{ rpm}$
- 4) Initial Rotor Angle: $\delta_0 = 0$
- 5) Source: Balanced 3 ϕ 380 V 50 Hz
- 6) Load Torque: $T_l = 0$

V. SIMULATION RESULTS

Figures 1–9 illustrate the responses obtained from MATLAB/Simulink simulation, for a time of 0.3s.

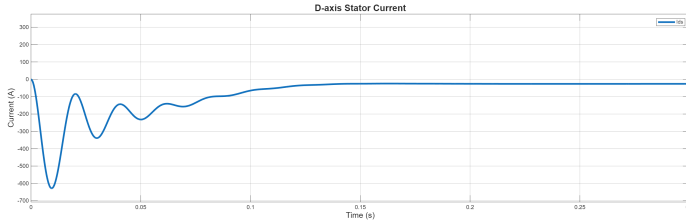


Fig. 1. Stator Current I_{ds} vs time

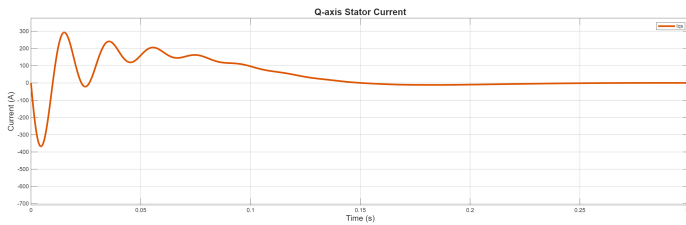


Fig. 2. Stator Current I_{qs} vs time

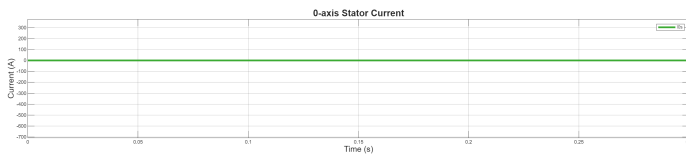


Fig. 3. Stator Current I_{0s} vs time

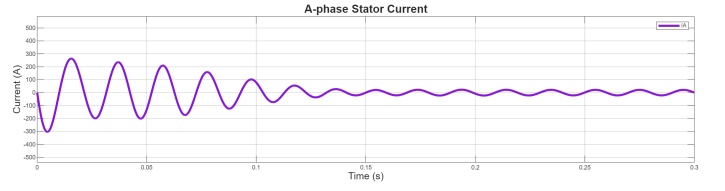


Fig. 4. Stator Current I_a vs time

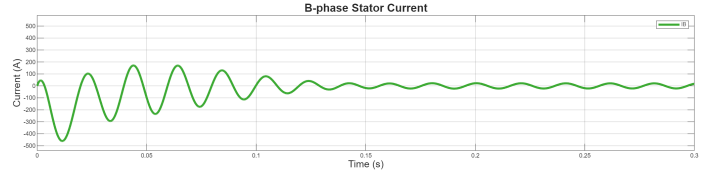


Fig. 5. Stator Current I_b vs time

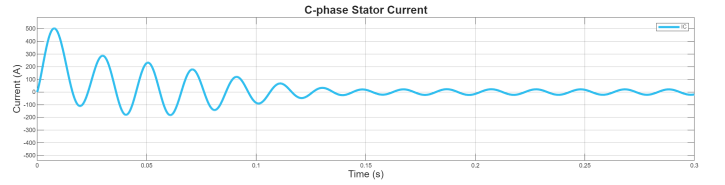


Fig. 6. Stator Current I_c vs time

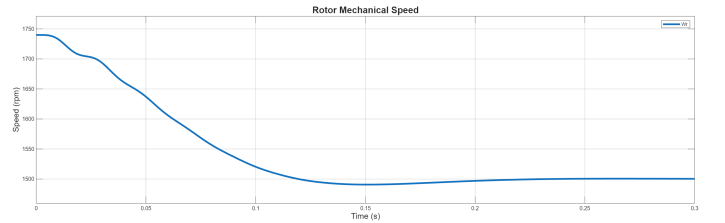


Fig. 7. Rotor Speed ω_r vs time

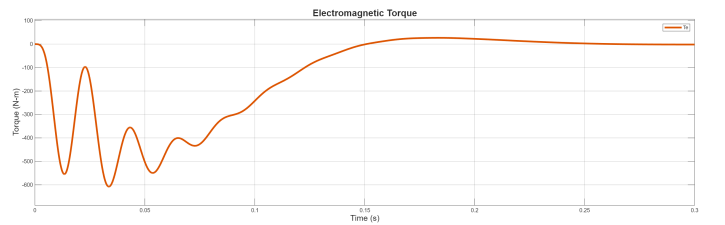


Fig. 8. Electromagnetic Torque T_e vs time

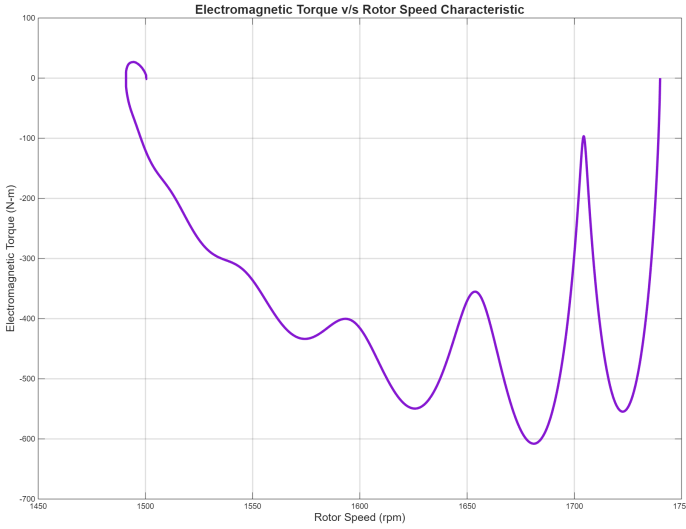


Fig. 9. Electromagnetic Torque T_e vs Rotor Speed ω_r .

VI. DISCUSSION

- 1) When the 3ϕ supply is applied to the motor running initially at 120% of its rated speed, a negative torque acts on the rotor and reduces its speed until it reaches synchronous speed (Fig. 7 and Fig. 8).
- 2) Since the friction and windage coefficient has been neglected, and the load torque assumed to be zero, the rotor speed settles to 1500 rpm instead of its rated speed of 1450 rpm.
- 3) The main advantage of using the synchronous reference frame to model the induction machine is that it represents the circuit voltages and currents into DC values. This is verified by the I_{ds} and I_{qs} plots (Fig. 1 and Fig. 2), which show that after the transient response, I_{ds} settles at a steady value of -26 A and I_{qs} converges to zero.
- 4) With a balanced supply and an isolated neutral, we can observe from the plot (Fig. 3) that I_{0s} current is zero.
- 5) With a balanced supply, the stator A, B, and C phase currents are also balanced after the transient period, and settle at a peak of 22 A , as is indicated by the plots (Fig. 4, Fig. 5, Fig. 6). This corresponds to the no-load current of the machine, when it supplies only the core losses.
- 6) When the machine initially has a speed higher than synchronous speed, it operates as a generator when connected to the supply. As the slip is negative, the currents and torque will also be negative. Since there is no load on the shaft to maintain this speed, the electromagnetic torque decelerates the rotor until it reaches synchronous speed. During this interval, the negative currents indicate that energy is fed back to the source. This is verified from the Torque-Speed Characteristic curve (Fig. 9) which shows a negative torque as the speed reduces from 1740 rpm to 1500 rpm.

VII. CONCLUSION

The initial transient behavior of the induction machine is accurately represented by the $(d-q)$ synchronous reference frame model. The simulation results show that at initial connection at super-synchronous speed, the machine operates in the generator region. The system experiences notable inrush currents and negative torque, both of which gradually settles to achieve a synchronous no-load operation.